

PCT 202 — Pharmaceutical Preparations & Calculations

Practice Workbook & Answer Key | ABUAD 200 Level Pharmacy | Second Semester

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How to use this workbook. The questions below are grouped by topic and were written from scratch by our team to help you revise the core concepts of PCT 202. They are original practice items for self-testing — not past exam papers and not a prediction of any future test. Try to answer each question on your own first, then check yourself against the answer key at the end. Always cross-check with your lecturer's current course outline.

Pharmaceutical Solutions, Solvents & Vehicles

1. A hydro-alcoholic elixir will generally dissolve a given drug more readily than a simple aqueous syrup. True or False?
2. When a prescription calls for distilled water, it is acceptable to supply purified water instead. True or False?
3. Rubbing (surgical) alcohol of the British Pharmacopoeia strength contains roughly 70% ethanol. True or False?
4. Glycerol mixes freely with water but is immiscible with alcohol. True or False?
5. Which single liquid is regarded as the chief solvent for most organic drug substances?
6. Which of the following is unsuitable as a solvent for oral (internal) preparations because of its toxicity: glycerol, propylene glycol, or ethylene glycol?
7. Water is sometimes employed as a co-solvent in a solvent blend. True or False?
8. A drug is always more soluble in one pure solvent than in a mixture of solvents. True or False?
9. Invert syrup is made up of which two simple sugars?
10. To prepare an aromatic (medicated) water by dilution, roughly how many volumes of water are added to one volume of the concentrate?
11. List five (5) advantages and five (5) disadvantages of formulating a medicine as a pharmaceutical solution.
12. Distinguish clearly between a saturated, an unsaturated, and a supersaturated solution.
13. Outline how a medicated elixir is prepared, give three (3) examples of medicated elixirs, and note the incompatibilities that may arise during elixir preparation.

Solubility, Dissolution & Co-solvency

14. Excipients are added to a formulation to control drug-release rate, appearance, taste, chemical stability and physical stability. True or False?

- 15.** When dissolution is endothermic (heat is absorbed), the solubility of the compound rises as temperature increases. True or False?
- 16.** When dissolution is exothermic (heat is evolved), solubility rises as temperature increases. True or False?
- 17.** The solubility of a gas in a liquid increases with a decrease in which factor?
- 18.** The solubility of a pharmaceutical solute depends on which of these: temperature, the solute, or the solvent?
- 19.** What name is given to the maximum mass of solute that can dissolve in 100 g of solvent at room temperature?
- 20.** The maximum quantity of solute dissolving in a fixed amount of a given solvent does NOT depend on which factor: temperature, pressure, nature of solute, or nature of solvent?
- 21.** Solubility is an intrinsic property of a substance; by which approach can it genuinely be altered?
- 22.** If the pH of a system is adjusted so that un-ionised molecules form in a concentration exceeding their solubility, what happens?
- 23.** The poor aqueous solubility of metronidazole can be increased several-fold by adding which co-solvent(s)?

Partition & Nernst's Distribution Law

- 24.** The partition coefficient can be used to gauge how efficiently one solvent extracts a compound dissolved in another. True or False?
- 25.** Which of the following is NOT an application of partitioning in pharmacy: miscibility separation, plant extraction, preservation of oil–water systems, or drug absorption and distribution in the body?
- 26.** The efficiency with which one solvent extracts a compound from another solvent is best measured by which quantity?
- 27.** In liquid–liquid extraction, the most efficient recovery of a compound is achieved by which technique?
- 28.** State five (5) applications of Nernst's distribution (partition) law in pharmacy.

Colligative Properties & Vapour Pressure

- 29.** The relationship linking the pH, the solubility, and the pK_a of an acidic drug is described by Henry's solubility law. True or False?
- 30.** Colligative properties arise from the chemical nature of the solute particles. True or False?
- 31.** As the mole fraction of the solvent in a solution becomes smaller, the vapour pressure of the solvent above the solution also becomes smaller. True or False?
- 32.** A solution normally has a lower vapour pressure than the pure solvent. True or False?
- 33.** Colligative behaviour is characteristic of which type of system: solutions, gases, suspensions or emulsions?

34. If the mole fraction of the solvent in a solution decreases, which of these increases: vapour pressure, boiling point, or osmotic pressure?
35. A sweetening solution prepared at 27 °C contains 90.1 g of fructose ($M = 180.2$) dissolved in 500 g of water ($M = 18.02$). The vapour pressure of pure water at 27 °C is 18.535 mmHg. Calculate (i) the relative lowering of vapour pressure, (ii) the lowering of the vapour pressure, and (iii) the vapour pressure of the solution.

The Phase Rule & Equilibrium

36. A system is at equilibrium when which form of energy is at a minimum (zero net change)?
37. In phase-rule terms, a phase is described as a ____ portion of a system.
38. For a phase to remain in equilibrium, which three variables must be held constant?
39. Phase equilibrium essentially describes what condition existing between phases when they coexist?
40. Applying the Gibbs phase rule, what is the degree of freedom (F) when all three phases of water coexist in equilibrium?
41. The condition in which the three phases of water coexist in equilibrium is known by what name?
42. By the Gibbs phase rule, what is the degree of freedom (F) when only two phases of water are in equilibrium?
43. A single pure compound can exist in three different phases. True or False?
44. In a one-component, two-phase system, temperature and pressure can be varied independently without changing the number of phases. True or False?
45. The greater the number of phases of a system in equilibrium, the greater the number of degrees of freedom. True or False?
46. Thermodynamic equilibrium is reached when the system attains which energy condition?
47. The phase rule is expressed mathematically by which equation?
48. Using the phase rule, what is the degree of freedom of a two-component, one-phase system?
49. Along the fusion (melting) curve of the water phase diagram, which phases are in equilibrium?
50. State the Gibbs phase-rule equation, apply it to a solution of salt in water, and name the variables that define the system.
51. Define a eutectic mixture and explain what it means for a system to be in equilibrium.

Freeze Drying (Lyophilisation)

52. What is freeze drying, and briefly describe the stages involved in the process.
53. Give four (4) disadvantages of freeze drying.
54. At what point during the process is the sublimation of water achieved?

Dosage Forms — Powders, Granules, Capsules, Tablets & Dispersed Systems

- 55. An indiffusible suspension is one that contains light, easily dispersed powders. True or False?
- 56. In a pharmaceutical suspension, all of the active ingredient is insoluble in the vehicle. True or False?
- 57. Excipients are needed in the formation of pharmaceutical granules. True or False?
- 58. Dusting powders are always non-medicated. True or False?
- 59. Medicated dusting powders may sometimes be adapted for oral (internal) use. True or False?
- 60. Low-density drugs are the easiest to compress into tablets. True or False?
- 61. Which of the following are pharmaceutical disperse systems: emulsions, lotions, suspensions — or all of these?
- 62. Which type of wrapping paper best protects a divided hygroscopic powder?
- 63. Which capsule size number has the smallest capacity?
- 64. By which principal method are most pharmaceutical tablets manufactured?
- 65. In a tabular form, differentiate between liniments and lotions with respect to viscosity, cosmetic appearance, application with friction, rate of absorption, and purpose/use.
- 66. Granules are solid, dry aggregates of powder particles strong enough to withstand handling, produced by a process called granulation. State whether this is true or false, and give four (4) reasons why granulation is important.
- 67. What are pharmaceutical tablets, and what are the merits and demerits of tablets as an oral dosage form?

Pharmaceutical Calculations

- 68. In pharmaceutical calculations, accuracy is very important because it is NOT directly related to patient health outcomes and safety. True or False?
- 69. In dilution calculations, the parameters of interest include the strength of the dilute solution, the strength of the concentrate, and the degree of dilution. True or False?
- 70. The ratio strength of a solution containing 1 mg of active ingredient in 0.5 mL is 1 in 5000. True or False?
- 71. A pharmacist dissolves 1.2 g of a medicinal agent in 60 mL of a cough syrup of specific gravity 1.20; the addition raises the volume by 0.2 mL. What is the specific gravity of the product (to 3 decimal places)?
- 72. How many milligrams of a 1:10 w/w powdered trituration of ibuprofen are needed to prepare 100 capsules for a clinical study if each capsule is to contain 0.5 mg of ibuprofen?
- 73. If 400 mL of a 20% w/v solution is diluted to 2 L, what is the final percentage strength?
- 74. A pharmacist must prepare 80 g of zinc oxide ointment at a concentration of 0.0002 mg per 2 g, using a stock ointment containing zinc oxide 0.025 g per 0.5 g. How much zinc oxide, and how much of the stock

ointment, should be used? (Solve by alligation and by proportionality.)

75. Rx: Benzalkonium chloride solution, 240 mL, to be made so that 10 mL diluted to one litre gives a 1:5000 w/v solution (Sig: apply as directed). How many litres of a 16% w/v stock solution of benzalkonium chloride are required to fill the prescription?

76. As part of a clinical study you must modify a standard 22 g pack of 2% w/w mupirocin ointment by adding mupirocin powder to raise it to 3% w/w. How many milligrams of mupirocin powder are required?

77. A pharmacist receives an order for 60 mL of a suspension containing cortisone acetate 1.5 mg/mL. She has a 360 mL bottle of suspension containing cortisone acetate 10 mg/5 mL and a diluent of sorbitol solution. Calculate the quantities of each ingredient used (by alligation and proportionality).

78. Rx: Sodium bicarbonate 15 g (solubility 1 in 11 of water), sodium chloride 10 g (solubility 1 in 3 of water), peppermint emulsion 25 mL, double-strength (D/S) chloroform water to 1000 mL. (i) Describe, with reasons, how you would dissolve the two solids. (ii) What minimum quantity of D/S chloroform water is needed to dissolve them if only 100 mL of the preparation is required, and — besides being a vehicle — what other functions does D/S chloroform water serve? (iii) At what point would you add the peppermint emulsion, and why?

Dispensing Practice & Tonicity

79. Briefly explain the relevance of e-prescription (electronic prescribing) in modern dispensing practice.

80. Differentiate between hypertonic, hypotonic and isotonic solutions, and describe what happens to a cell placed in each.

Answer Key

Worked answers, grouped to match the questions above.

Pharmaceutical Solutions, Solvents & Vehicles

1. True. Because an elixir is a clear hydro-alcoholic vehicle, it dissolves both water-soluble and alcohol-soluble drugs, so many actives are more soluble in it than in a purely aqueous syrup.
2. False. Purified water and distilled water are not interchangeable on demand — distilled water should be dispensed when specifically ordered, as the two are prepared and specified differently.
3. True. Surgical/rubbing alcohol is approximately 70% v/v ethanol (about 90 parts of ethanol diluted with water), the concentration giving best antiseptic action.
4. False. Glycerol is miscible with BOTH water and alcohol; it is only immiscible with fixed oils.
5. Alcohol (ethanol). It is the principal solvent for organic compounds that are poorly soluble in water.
6. Ethylene glycol. It is toxic and must never be used internally; glycerol and propylene glycol are safe internal solvents.
7. True. Water frequently forms part of a co-solvent system (e.g. water–alcohol–glycerol) used to raise the solubility of a drug.
8. False. Solubility can often be markedly increased by using a co-solvent blend rather than a single solvent (the co-solvency principle).
9. Glucose and fructose. Invert syrup is produced by hydrolysing sucrose into its two monosaccharides, glucose and fructose.
10. About 49 volumes. A 2% concentrate is diluted 1 in 50 — one volume of concentrate to forty-nine volumes of water.
11. **Advantages:** (1) drug is already in solution, giving rapid and uniform absorption; (2) dose is easy to adjust and measure; (3) no need to shake, so dosing is consistent; (4) easier to swallow than solids, useful for children and the elderly; (5) can mask taste and buffer irritant drugs. **Disadvantages:** (1) bulky and awkward to transport/store; (2) generally less chemically and physically stable than solids; (3) prone to microbial growth, needing preservatives; (4) unpleasant tastes are harder to hide fully; (5) measuring errors by the patient and spillage are common.
12. A **saturated** solution holds the maximum solute that will dissolve at a given temperature, in equilibrium with undissolved solute. An **unsaturated** solution holds less than this maximum, so more solute could still dissolve. A **supersaturated** solution transiently holds more than the equilibrium amount (usually by cooling a hot saturated solution) and is unstable — the excess crystallises out on disturbance.
13. **Preparation:** the drug is dissolved in the alcoholic portion, water-soluble components in the aqueous portion, the two are combined (adding the aqueous phase to the alcoholic phase, with stirring), flavour/sweetener added, then made to volume and clarified by filtration. **Examples:** paracetamol elixir, diphenhydramine (Benadryl) elixir, phenobarbital elixir. **Incompatibilities:** water-soluble drugs may precipitate on adding alcohol (and vice-versa); high sugar or salt content can salt-out the alcohol-soluble components; and adding water too quickly to the alcoholic phase causes cloudiness/separation.

Solubility, Dissolution & Co-solvency

- 14.** False. Excipients are chosen to improve chemical STABILITY, not instability — the statement wrongly lists 'chemical instability' as a goal.
- 15.** True. For an endothermic dissolution, raising the temperature favours dissolving, so solubility increases with heat.
- 16.** False. For an exothermic dissolution, raising the temperature opposes dissolving, so solubility DECREASES with heat.
- 17.** Temperature. Gases are more soluble at lower temperatures (and higher pressures).
- 18.** All of them. Solubility depends jointly on temperature, the nature of the solute, and the nature of the solvent.
- 19.** Solubility. (Strictly, the equilibrium solubility expressed per 100 g of solvent.)
- 20.** Pressure. For solids/liquids in a liquid, pressure has negligible effect; temperature and the nature of solute and solvent are what matter.
- 21.** By modifying the chemical structure of the molecule. Changing molecular weight alone or swapping hydrophilic for hydrophobic groups does not reliably raise solubility.
- 22.** Precipitation occurs. The un-ionised form exceeds its solubility limit and comes out of solution.
- 23.** All of the above — ethanol, propylene glycol and glycerin all act as co-solvents that raise its solubility.

Partition & Nernst's Distribution Law

- 24.** True. The partition coefficient expresses how a solute distributes between two immiscible solvents, so it directly reflects extraction efficiency.
- 25.** Miscibility separation. Partitioning underlies plant extraction, oil–water preservative distribution, and drug absorption/distribution — but 'miscibility separation' is not one of its applications.
- 26.** The partition coefficient — the equilibrium ratio of the solute's concentrations in the two solvents.
- 27.** Using several small portions of the extracting solvent in successive (multiple) extractions, rather than one large portion at once.
- 28.** (1) Solvent (liquid–liquid) extraction of drugs and phytochemicals; (2) prediction of drug absorption and distribution across biological membranes; (3) determining the distribution of a preservative between the oil and water phases of an emulsion; (4) understanding drug partitioning into tissues and protein binding; (5) analytical assays and purification steps that separate compounds between two phases.

Colligative Properties & Vapour Pressure

- 29.** False. That relationship is given by the Henderson–Hasselbalch equation, not by Henry's law (which relates gas solubility to partial pressure).
- 30.** False. Colligative properties depend on the NUMBER of solute particles present, not on their chemical nature.
- 31.** True. By Raoult's law the solvent's vapour pressure is proportional to its mole fraction, so a smaller solvent mole fraction means a lower vapour pressure.
- 32.** True. Dissolved (non-volatile) solute lowers the solvent's vapour pressure below that of the pure solvent.

- 33.** Solutions. Colligative properties (vapour-pressure lowering, boiling-point elevation, freezing-point depression, osmotic pressure) are properties of solutions.
- 34.** Osmotic pressure increases (and boiling point also rises); note that the vapour pressure of the solution FALLS rather than rises, so the vapour-pressure option is incorrect.
- 35.** Moles fructose = $90.1 / 180.2 = 0.500$ mol; moles water = $500 / 18.02 = 27.75$ mol. **(i)** Relative lowering = mole fraction of solute = $0.500 / (0.500 + 27.75) = 0.500 / 28.25 = 0.0177$. **(ii)** Lowering $\Delta p = 0.0177 \times 18.535 = 0.328$ mmHg. **(iii)** Vapour pressure of solution = $18.535 - 0.328 = 18.207$ mmHg.

The Phase Rule & Equilibrium

- 36.** The free energy. At equilibrium the change in Gibbs free energy is zero (free energy is at a minimum).
- 37.** A homogeneous portion — a phase is any physically and chemically homogeneous, mechanically separable part of a system.
- 38.** Temperature, pressure and volume must remain constant.
- 39.** A balance (equilibrium balance) between the coexisting phases, with no net transfer of matter between them.
- 40.** $F = 0$. Three phases of a one-component system gives an invariant point ($F = 1 - 3 + 2 = 0$).
- 41.** The triple point — the unique temperature and pressure at which solid, liquid and vapour coexist.
- 42.** $F = 1$. Two phases of a one-component system leave one degree of freedom ($F = 1 - 2 + 2 = 1$).
- 43.** True. A pure substance can exist as solid, liquid and vapour (three phases).
- 44.** False. A one-component two-phase system has $F = 1$, so only ONE variable can be varied independently while both phases persist.
- 45.** False. The opposite is true — increasing the number of phases DECREASES the degrees of freedom ($F = C - P + 2$).
- 46.** The overall LOWEST energy state of the system — the point of minimum free energy.
- 47.** The Gibbs equation (the Gibbs phase rule, $F = C - P + 2$).
- 48.** $F = C - P + 2 = 2 - 1 + 2 = 3$.
- 49.** Ice and water (solid and liquid) — the fusion curve is the solid–liquid equilibrium line.
- 50. Equation:** $F = C - P + 2$, where C = number of components, P = number of phases, F = degrees of freedom. **Applied to salt + water:** $C = 2$. For a single (liquid) solution phase, $F = 2 - 1 + 2 = 3$; for the solution in equilibrium with solid salt (two phases), $F = 2 - 2 + 2 = 2$. **Defining variables:** temperature, pressure and composition (concentration).
- 51.** A **eutectic mixture** is a mixture of two or more solids in the exact proportion that melts (and solidifies) at a single, lowest temperature — lower than either pure component. A system is in **equilibrium** when its phases coexist with no net change over time, i.e. the rates of opposing processes are equal and the free energy is at a minimum.

Freeze Drying (Lyophilisation)

- 52. Freeze drying (lyophilisation)** is the removal of water from a frozen product by sublimation under reduced pressure, leaving a dry, porous cake. **Stages:** (1) *freezing* — the product is cooled until the water forms ice; (2)

primary drying — pressure is lowered and gentle heat applied so the ice sublimates directly to vapour; (3)
secondary drying — residual bound (unfrozen) moisture is desorbed at a slightly higher temperature to give a low final moisture content.

53. (1) It is slow and time-consuming; (2) it is expensive, needing specialised vacuum/refrigeration equipment; (3) the process is energy-intensive with high running costs; (4) the porous product is fragile/hygroscopic and must be sealed carefully to keep out moisture.

54. During primary drying — sublimation occurs when the frozen water is held below the triple point (low temperature under reduced pressure), so ice passes directly to vapour without melting.

Dosage Forms — Powders, Granules, Capsules, Tablets & Dispersed Systems

55. False. Indiffusible suspensions contain heavy, dense powders that settle rapidly and do not disperse readily; light, easily dispersed powders give diffusible suspensions.

56. True. The active in a true pharmaceutical suspension is insoluble (or only very slightly soluble) in the vehicle, which is why it is suspended rather than dissolved.

57. True. Binders, diluents, disintegrants and other excipients are required to form good granules.

58. False. Dusting powders may be either medicated or non-medicated.

59. False. Dusting powders are intended for external application to the skin and are not adapted for oral use.

60. False. Low-density (bulky) drugs are generally difficult to compress; they tend to cap or laminate and often need granulation first.

61. All of the above — emulsions, lotions and suspensions are all disperse systems.

62. Waxed paper — its water-repellent surface best protects a hygroscopic powder from moisture.

63. Size 5 — capsule sizes run from 000 (largest) down to 5 (smallest), so 5 has the least capacity.

64. Compression — the great majority of tablets are made by compressing powders or granules in a tablet press.

65. Liniments vs lotions:

Parameter	Liniments	Lotions
Viscosity	More viscous (oily/alcoholic)	Less viscous, more fluid
Cosmetic appearance	Oily, greasy on the skin	Non-greasy, cosmetically elegant
Application of friction	Applied WITH rubbing/massage	Applied WITHOUT friction (dabbed on)
Rate of absorption	Slower, prolonged local action	Faster absorption / evaporation
Purpose / use	Counter-irritant, rubefacient, analgesic	Soothing, cooling, protective, antiseptic

66. True. Granules are indeed solid, dry aggregates made by granulation. **Why granulation matters:** (1) it improves powder flow, giving uniform die filling and consistent tablet/capsule weight; (2) it improves compressibility so tablets form properly; (3) it prevents segregation of a powder mix, keeping the blend uniform; (4) it reduces dust and improves handling, and can control dissolution/wetting of the drug.

67. Pharmaceutical tablets are solid dosage forms containing a drug with excipients, usually made by compressing a powder or granulation into a defined shape and size. **Merits:** accurate unit dosing, good chemical/physical stability, convenient and portable, easy to mass-produce cheaply, tasteless/odour-masked,

and amenable to modified release. **Demerits:** difficult to swallow for children and the elderly, unsuitable for drugs with poor compressibility or very large doses, slower onset than solutions, and some drugs have poor bioavailability from tablets.

Pharmaceutical Calculations

68. False. Accuracy is directly related to patient safety and health outcomes — an error changes the delivered dose and can harm the patient.

69. True. Dilution problems are governed by the concentrate strength, the final (dilute) strength, and the degree/factor of dilution.

70. False. $1 \text{ mg in } 0.5 \text{ mL} = 2 \text{ mg per mL} = 1 \text{ g per } 500 \text{ mL} = \mathbf{1 \text{ in } 500}$, not 1 in 5000.

71. Mass of syrup = $60 \times 1.20 = 72 \text{ g}$; total mass = $72 + 1.2 = 73.2 \text{ g}$; final volume = 60.2 mL . SG = $73.2 / 60.2 = \mathbf{1.216}$.

72. Each capsule needs 0.5 mg , so 100 capsules need 50 mg of ibuprofen. A 1:10 trituration contains 1 part drug in 10 parts, so mass required = $50 \times 10 = \mathbf{500 \text{ mg}}$ of the trituration.

73. Using $C_1V_1 = C_2V_2$: $20\% \times 400 \text{ mL} = C_2 \times 2000 \text{ mL}$, so $C_2 = 8000 / 2000 = \mathbf{4\% \text{ w/v}}$.

74. Target = $0.0002 \text{ mg} / 2 \text{ g} = 0.0001 \text{ mg/g}$, so in 80 g the ointment must contain $0.0001 \times 80 = \mathbf{0.008 \text{ mg}}$ of zinc oxide. The stock is $0.025 \text{ g} / 0.5 \text{ g} = 0.05 \text{ g/g}$ (50 mg/g), so the mass of stock supplying $0.008 \text{ mg} = 0.008 / 50 = 0.00016 \text{ g} = \mathbf{0.16 \text{ mg of the stock ointment}}$, made up to 80 g with ointment base. (The very small quantities show that geometric dilution / trituration is needed in practice.)

75. For 1:5000 w/v, 1 L (1000 mL) of the final dilution contains 0.2 g ; that 0.2 g must come from the 10 mL taken from our preparation, so our 240 mL preparation must be $0.2 \text{ g} / 10 \text{ mL} = 2\% \text{ w/v}$. Total benzalkonium needed = $240 \times 0.02 = 4.8 \text{ g}$. From a 16% (0.16 g/mL) stock: volume = $4.8 / 0.16 = 30 \text{ mL} = \mathbf{0.03 \text{ L}}$.

76. The 22 g of 2% ointment contains 0.44 g mupirocin. Adding $x \text{ g}$ of pure mupirocin: $(0.44 + x) / (22 + x) = 0.03 \Rightarrow 0.44 + x = 0.66 + 0.03x \Rightarrow 0.97x = 0.22 \Rightarrow x = 0.2268 \text{ g} \approx \mathbf{226.8 \text{ mg} (\approx 227 \text{ mg})}$ of mupirocin powder.

77. Cortisone acetate needed = $60 \text{ mL} \times 1.5 \text{ mg/mL} = 90 \text{ mg}$. The stock is $10 \text{ mg} / 5 \text{ mL} = 2 \text{ mg/mL}$, so volume of stock = $90 / 2 = \mathbf{45 \text{ mL}}$; the balance, $\mathbf{15 \text{ mL of sorbitol solution}}$ ($60 - 45$), makes up the 60 mL .

78. (i) Dissolve each solid separately in the minimum water it needs, using its stated solubility: sodium bicarbonate $15 \text{ g} \times 11 = 165 \text{ mL}$, sodium chloride $10 \text{ g} \times 3 = 30 \text{ mL}$ — this guarantees each dissolves completely before combining and before making to volume. **(ii)** For 100 mL the quantities scale to one-tenth: sodium bicarbonate 1.5 g (needs $1.5 \times 11 = 16.5 \text{ mL}$) and sodium chloride 1 g (needs $1 \times 3 = 3 \text{ mL}$), so the minimum D/S chloroform water to dissolve them $\approx \mathbf{19.5 \text{ mL}}$. Besides being a vehicle, D/S chloroform water also acts as a **preservative (antimicrobial)** and as a **flavouring/sweetening and carminative** agent. **(iii)** The peppermint emulsion is added near the END, after the solids are dissolved and before making to final volume, because it is a volatile flavouring — adding it last minimises loss of the volatile oil and keeps the flavour in the finished product.

Dispensing Practice & Tonicity

79. E-prescriptions reduce errors from illegible handwriting, speed up dispensing and transmission between prescriber and pharmacy, improve record-keeping and auditing, allow automatic checks for dose errors, allergies and drug interactions, cut prescription fraud/loss, and improve overall patient safety and workflow efficiency.

80. A **hypertonic** solution has a higher solute concentration (osmotic pressure) than the cell contents; a cell placed in it loses water and shrinks (crenation in red cells). A **hypotonic** solution is more dilute than the cell;

water enters the cell, which swells and may burst (haemolysis of red cells). An **isotonic** solution has the same osmotic pressure as the cell contents; there is no net water movement and the cell keeps its normal shape and volume.

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